Raj Reddy

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Education

University of Utah **Salt Lake City, UT**
B.S. Computer Science *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies **Dallas, TX**
Associate Software Engineer *June 2025 – Present*

- Developed real-time signal processing algorithms in Python and C++ on embedded systems, applying DSP techniques for defense sensor data streams.
- Built microservices for high-frequency data ingestion and storage, leveraging Podman and serverless infrastructure on AWS.
- Automated CI/CD pipelines with static code analysis and integration testing.

Doxy.me **Charleston, SC**
Software Engineer *August 2024 – May 2025*

- Deployed and optimized transformer-based GenAI models (PyTorch, Llama) to production on AWS SageMaker and Fargate, implementing inference pipelines that processed multimodal inputs from WebRTC video sessions and delivered structured clinical outputs at scale.
- Improved model accuracy and reliability by integrating reinforcement learning into the clinical inference loop, evaluating model performance across diverse input distributions, and iterating on data processing and post-processing pipelines to reduce error rates in production.
- Built TypeScript service APIs and CI/CD automation that connected ML infrastructure to user-facing provider dashboards, enabling end-to-end model deployment from training through production monitoring with structured observability.

University of Utah **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.